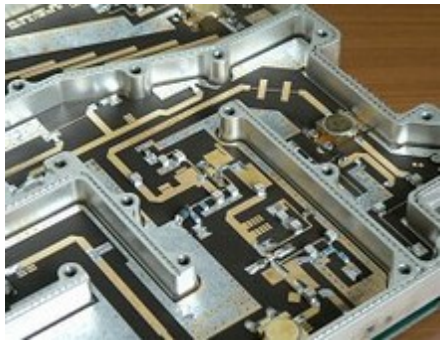
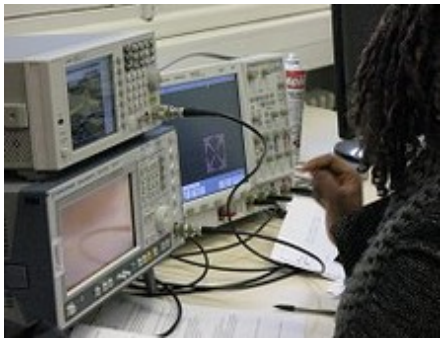
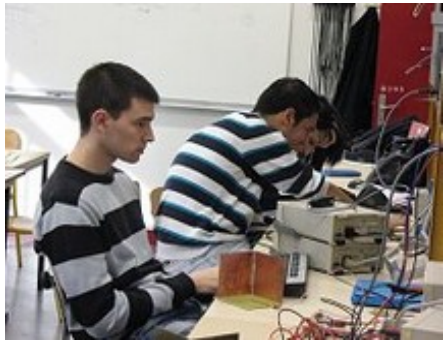
Notions sur les radars
Aspect pratique







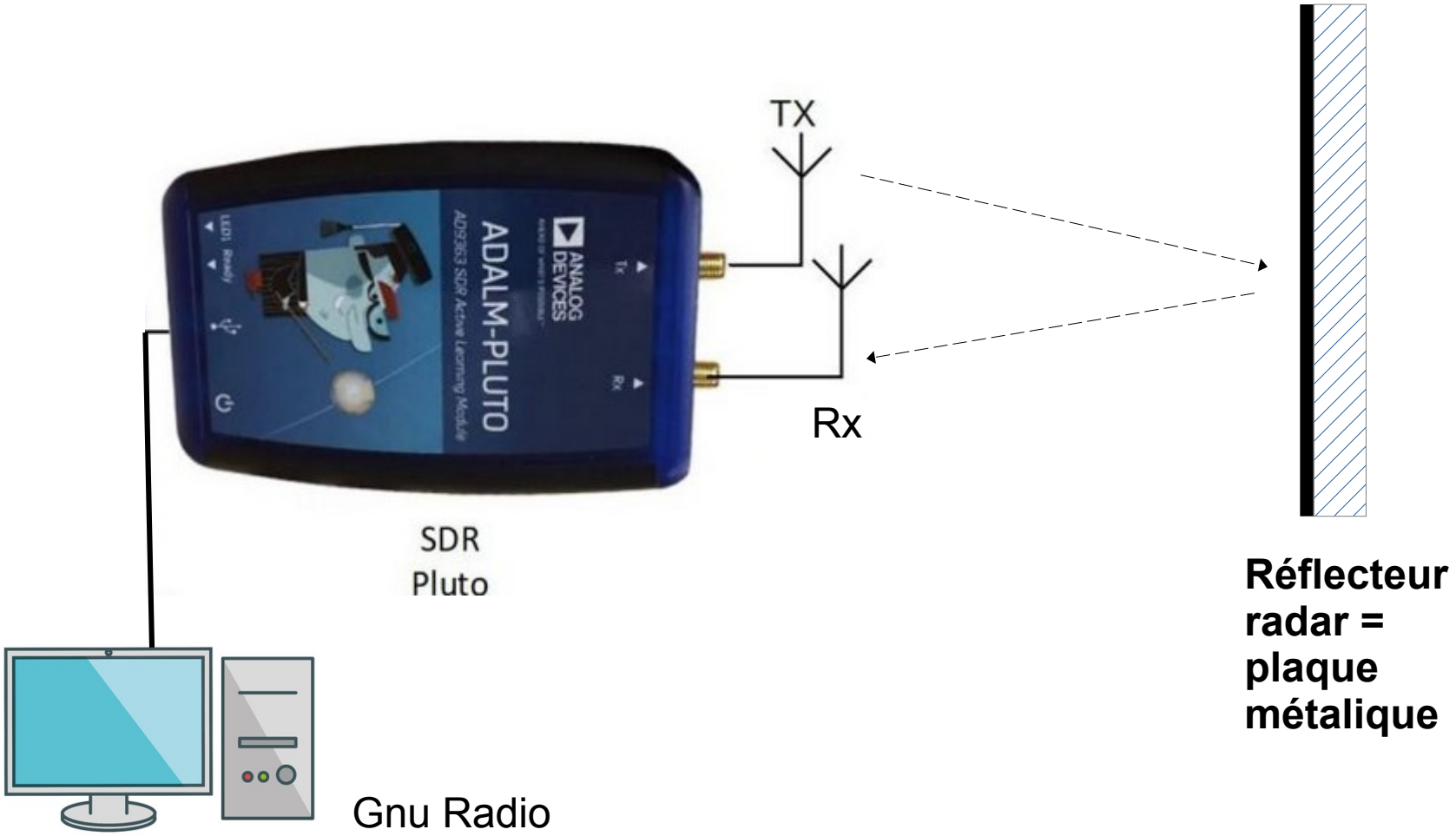
Simulateur de cible Radar

Franck Daout
fdaout@parisnanterre.fr

https://cva-geii.parisnanterre.fr/

CFD - Bourges

Notre système radar « low cost »

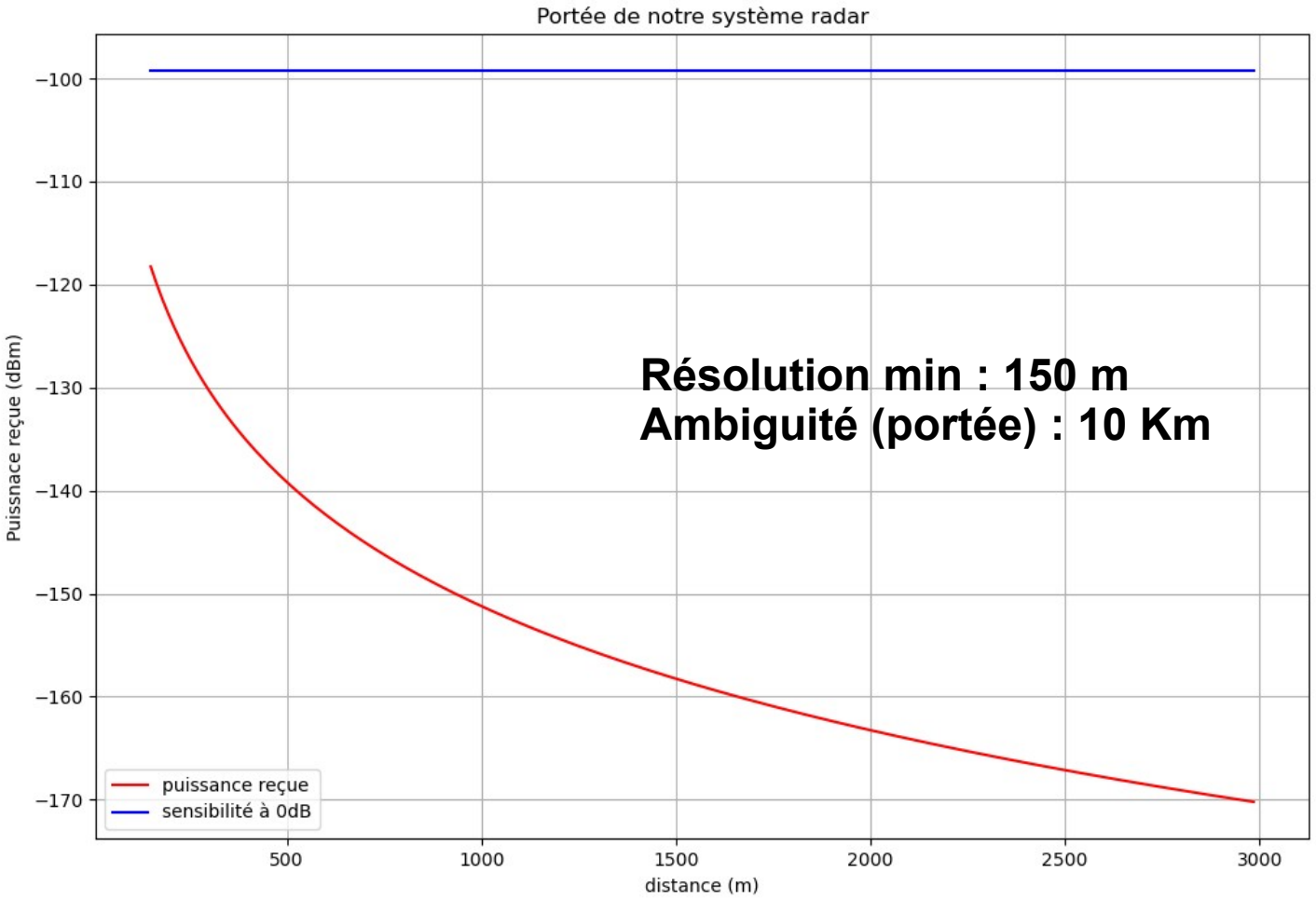


Caractéristiques de notre système radar

- Fréquence : 2450MHz
- Fréquence échantillonnage : 3MHz
- Largeur impulsion : 1 us ..50 us
- PRF : 15 KHz
- Puissance max. < 10dBm
- Gain antenne : 5 dBi
- Surface équivalente radar : 0 dBm2
- NF récepteur : 10 dB

Présentation

Valeurs quantitatives :

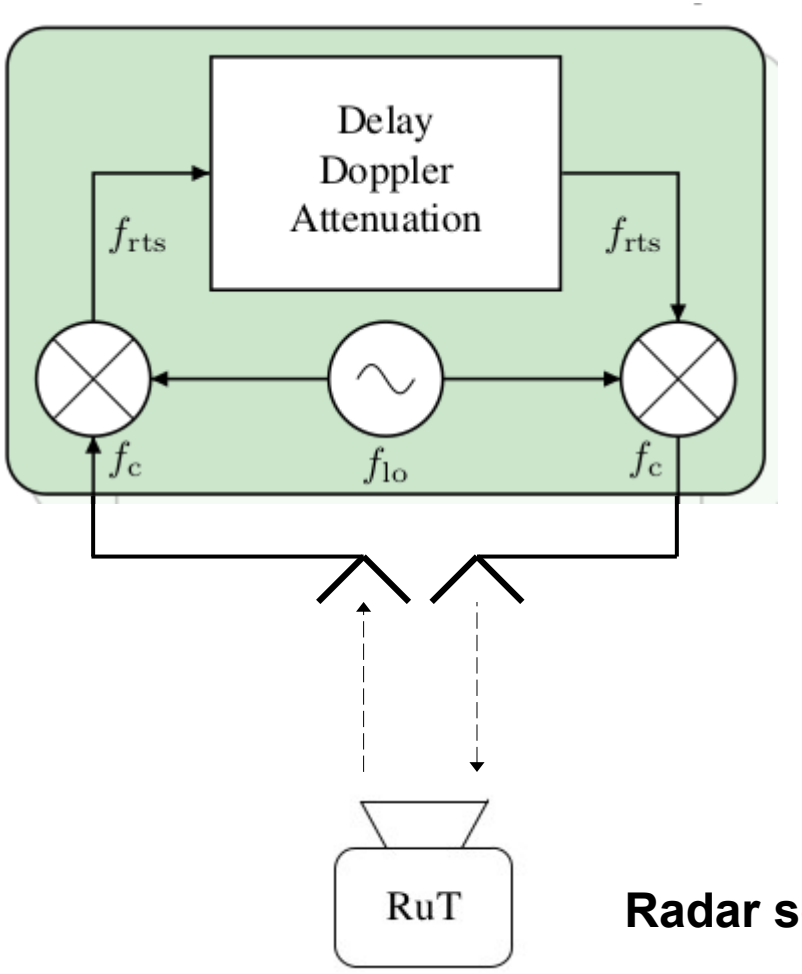


Détection impossible !

Solutions ? :

- Augmenter la puissance d'émission ?
- Diminuer la sensibilité du récepteur ?
- Modifier les antennes ?
- Changer la fréquence ?
- Augmenter la SER de la cible ?
- Diminuer la durée d'impulsion ?
- Utiliser un simulateur de cible ?

C'est quoi un simulateur de cible ….



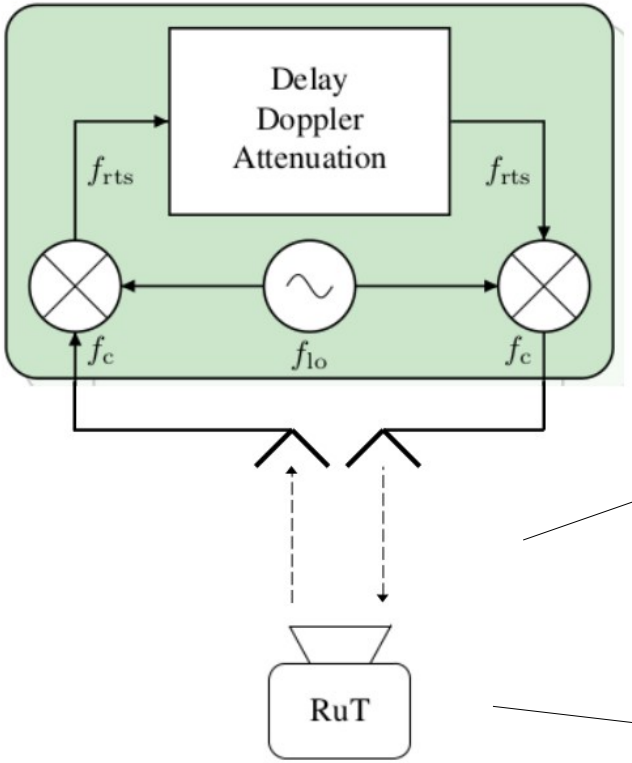
Simulateur de cibles

- Retard → position
- Attenuation → SER
- Doppler → vitesse

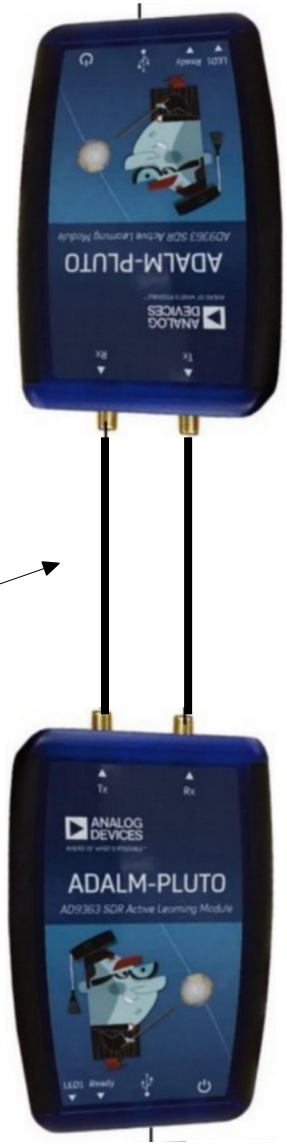
Radar sous test

Simulateur de cibles

Notre simulateur de cible « low cost »



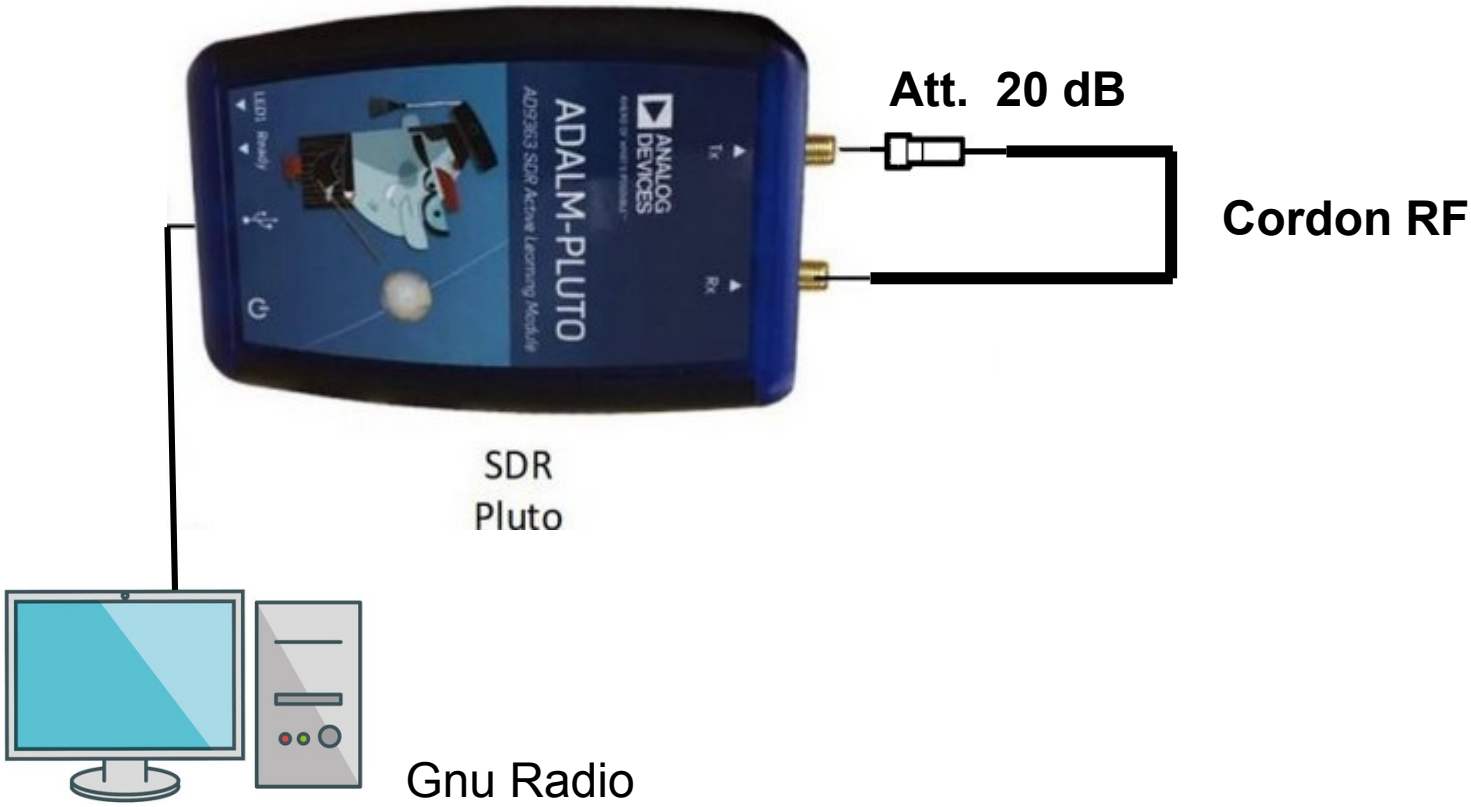
Simulateur de cibles



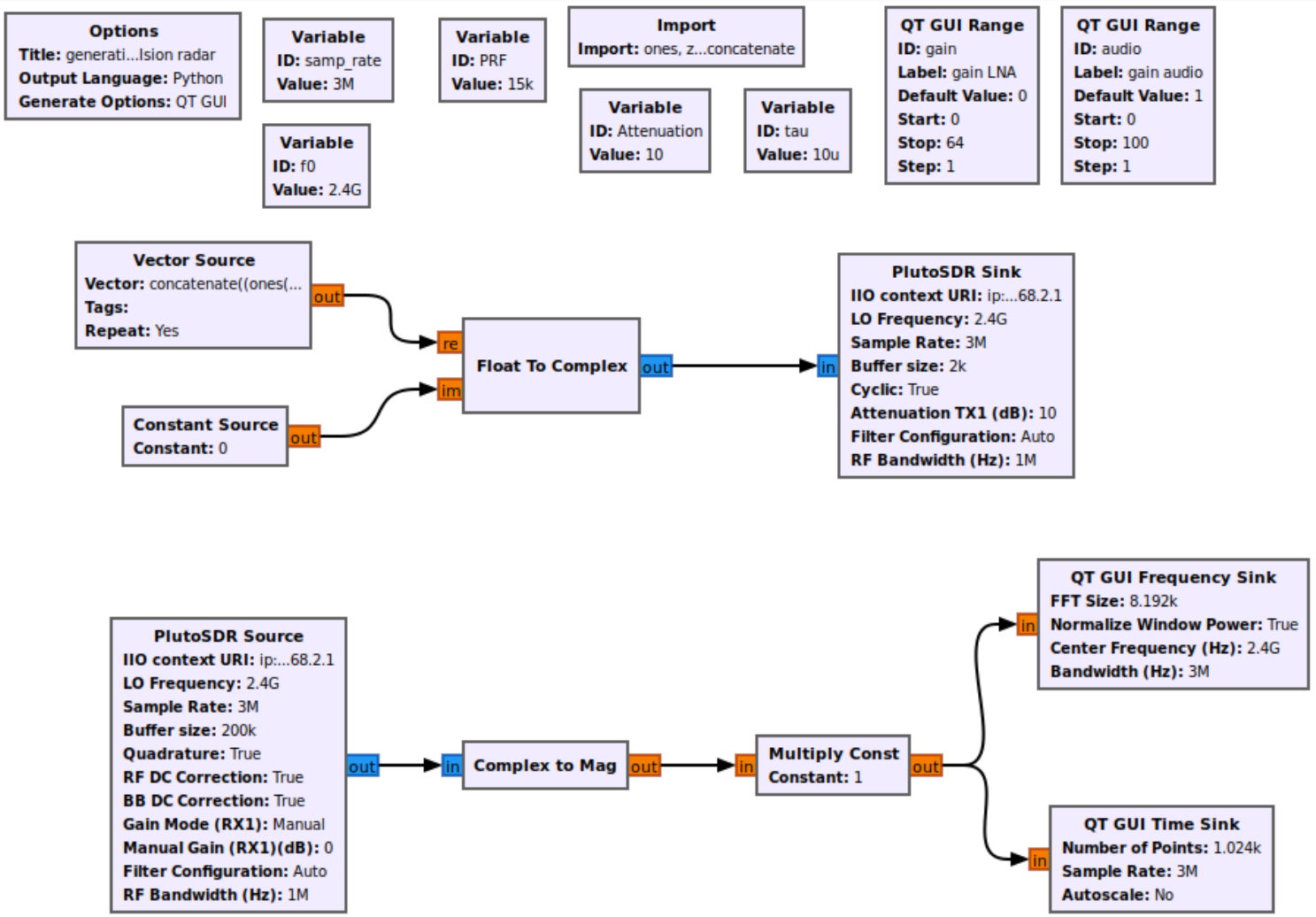
Canal de propagation

Radar sous test

Test de la partie radar

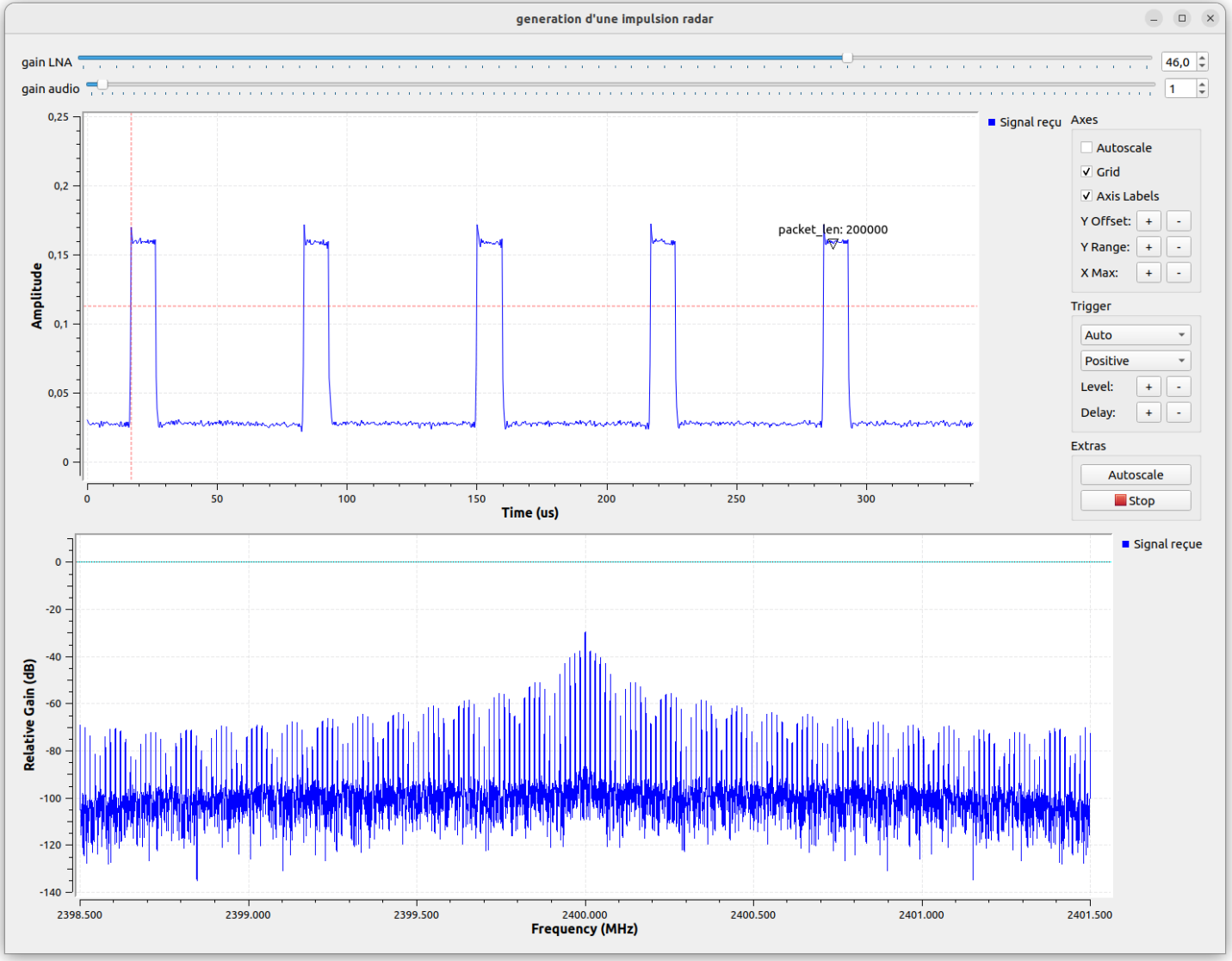


Notre Radar


Options
Title: generati...lsion radar
Output Language: Python
Generate Options: QT GUI
Variable
ID: samp_rate
Value: 3M
Variable
ID: f0
Value: 2.4G
Variable
ID: PRF
Value: 15k
Import
Import: ones, z...concatenate
Variable
ID: Attenuation
Value: 10
Variable
ID: tau
Value: 10u
QT GUI Range
ID: gain
Label: gain LNA
Default Value: 0
Start: 0
Stop: 64
Step: 1
QT GUI Range
ID: audio
Label: gain audio
Default Value: 1
Start: 0
Stop: 100
Step: 1
Vector Source
Vector: concatenate((ones(...
Tags:
Repeat: Yes
out
Constant Source
Constant: 0
out
re
im
Float To Complex
out
PlutoSDR Sink
IIO context URI: ip:...68.2.1
LO Frequency: 2.4G
Sample Rate: 3M
Buffer size: 2k
Cyclic: True
Attenuation TX1 (dB): 10
Filter Configuration: Auto
RF Bandwidth (Hz): 1M
in
PlutoSDR Source
IIO context URI: ip:...68.2.1
LO Frequency: 2.4G
Sample Rate: 3M
Buffer size: 200k
Quadrature: True
RF DC Correction: True
BB DC Correction: True
Gain Mode (RX1): Manual
Manual Gain (RX1)(dB): 0
Filter Configuration: Auto
RF Bandwidth (Hz): 1M
out
in
Complex to Mag
out
in
Multiply Const
Constant: 1
out
in
QT GUI Frequency Sink
FFT Size: 8.192k
Normalize Window Power: True
Center Frequency (Hz): 2.4G
Bandwidth (Hz): 3M
in
QT GUI Time Sink
Number of Points: 1.024k
Sample Rate: 3M
Autoscale: No

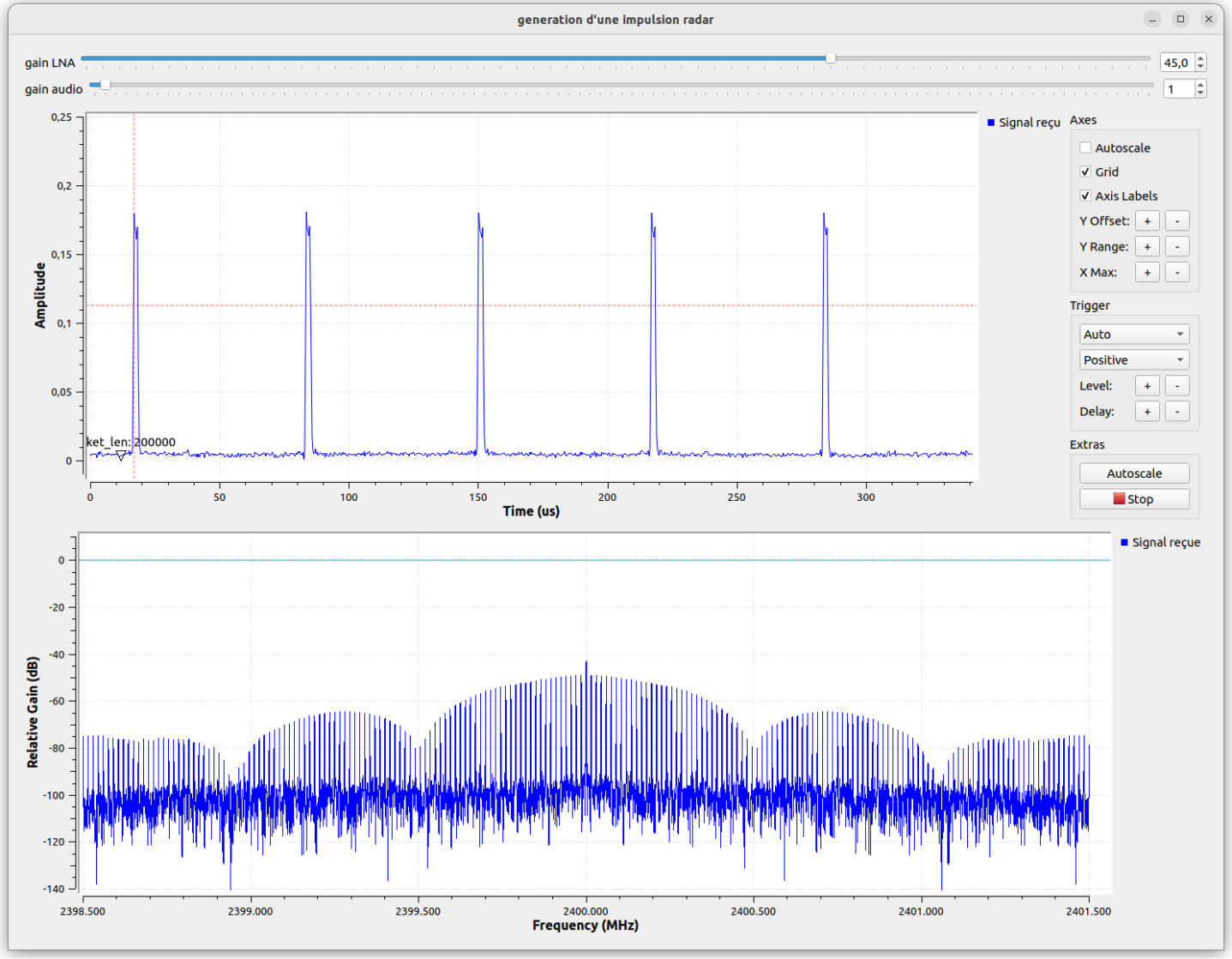
Notre Radar

Impulsion 10us

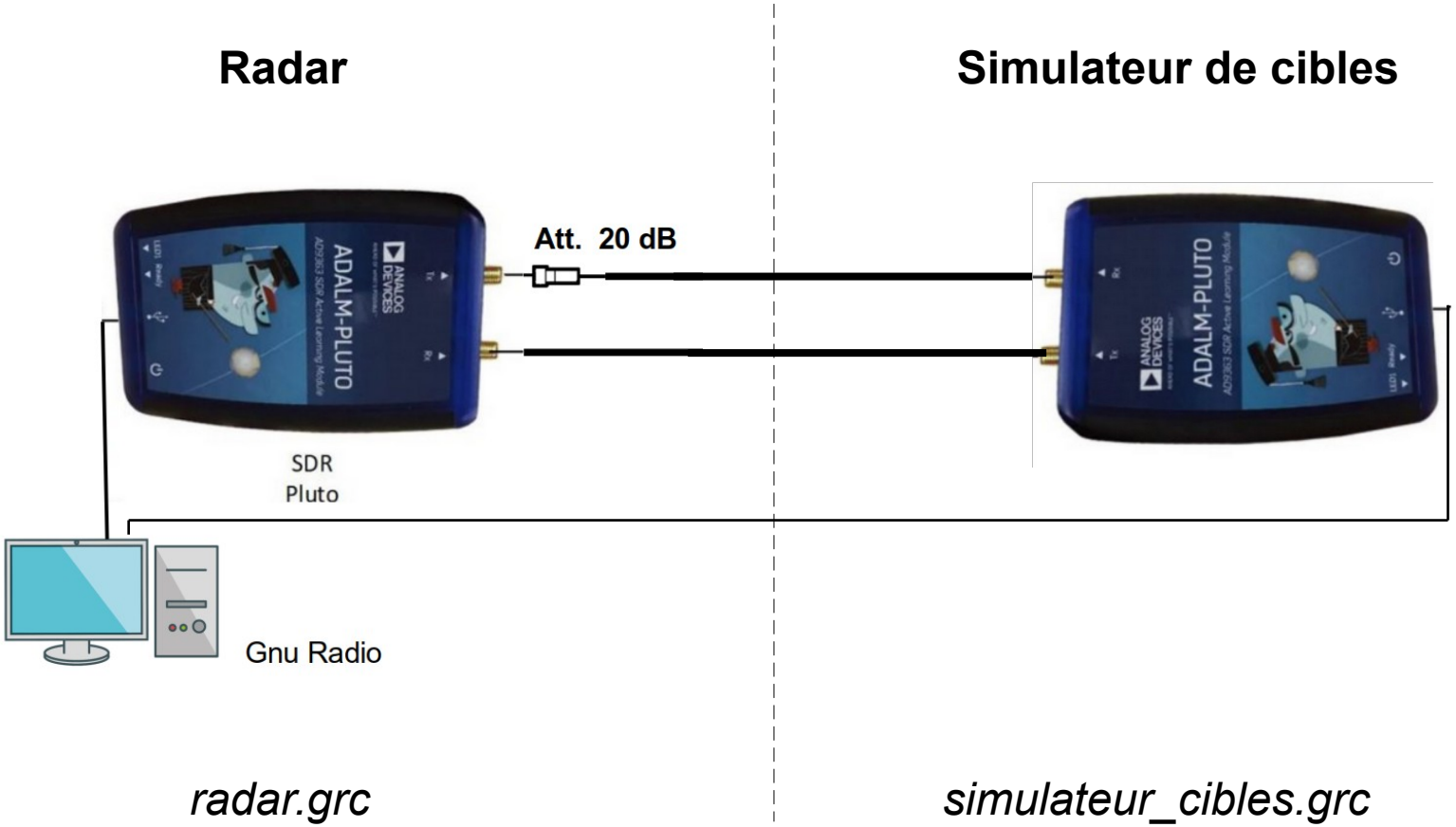


Notre Radar

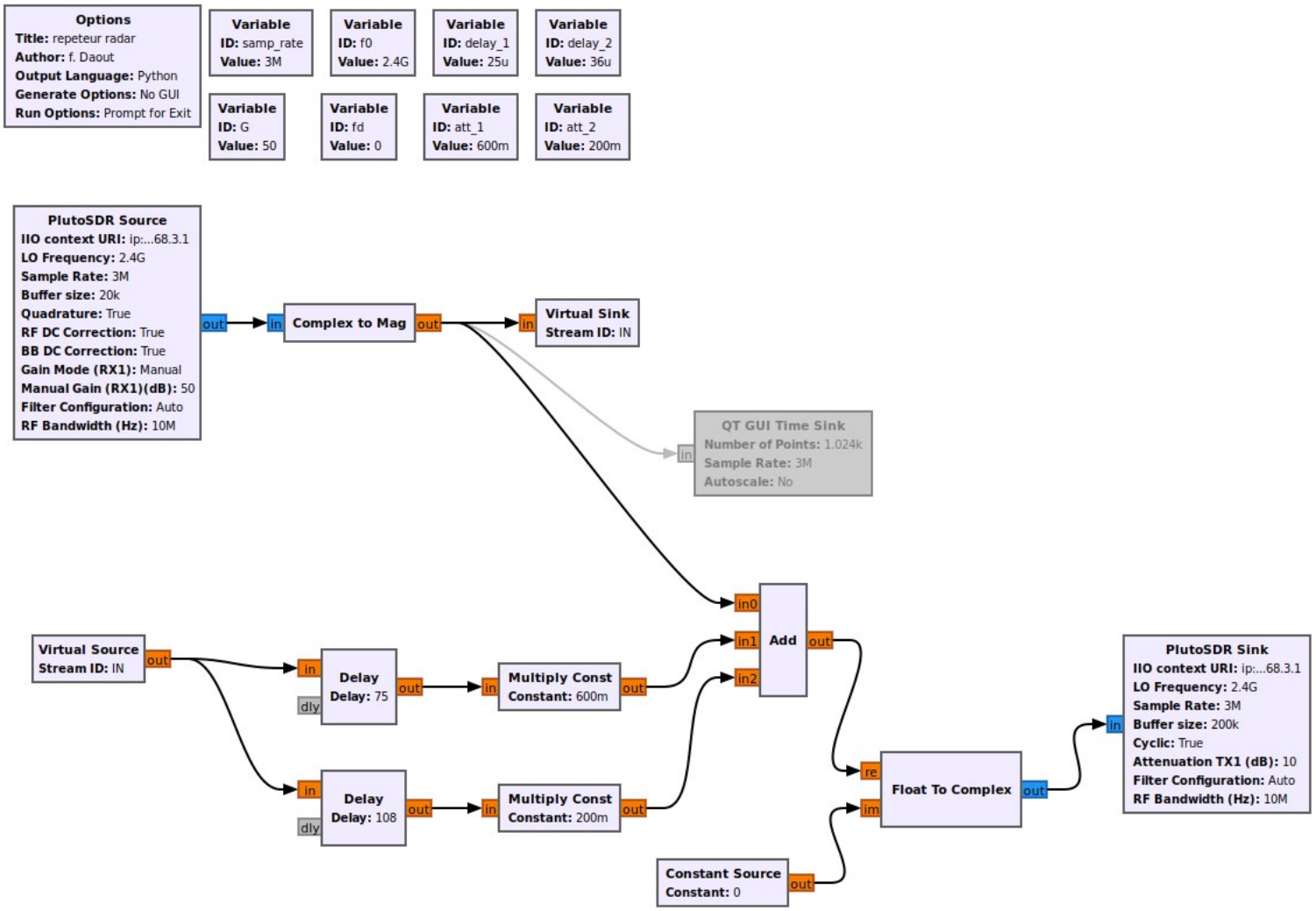
Impulsion 2us



Test du simulateur de cibles

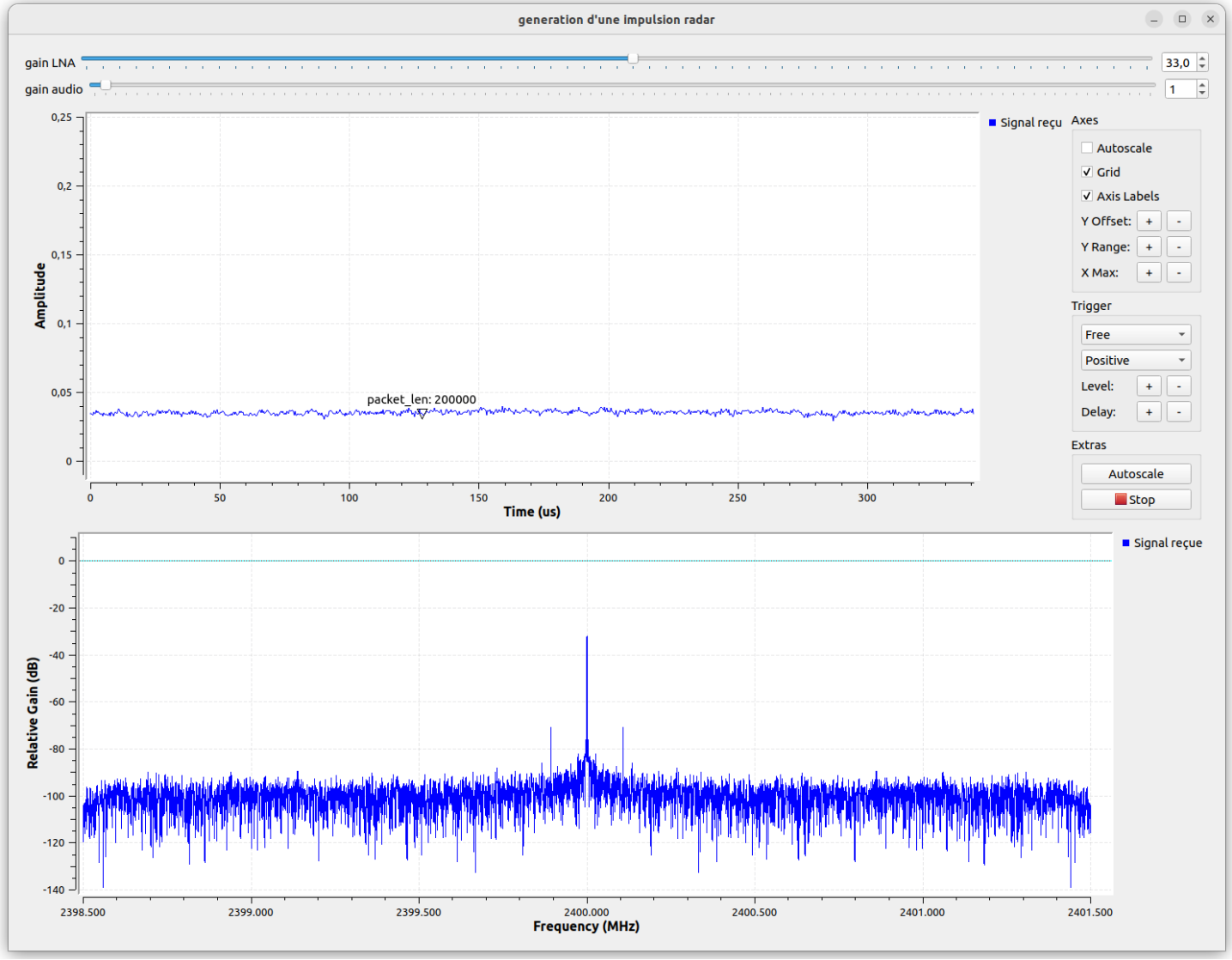


Simulateur de cibles



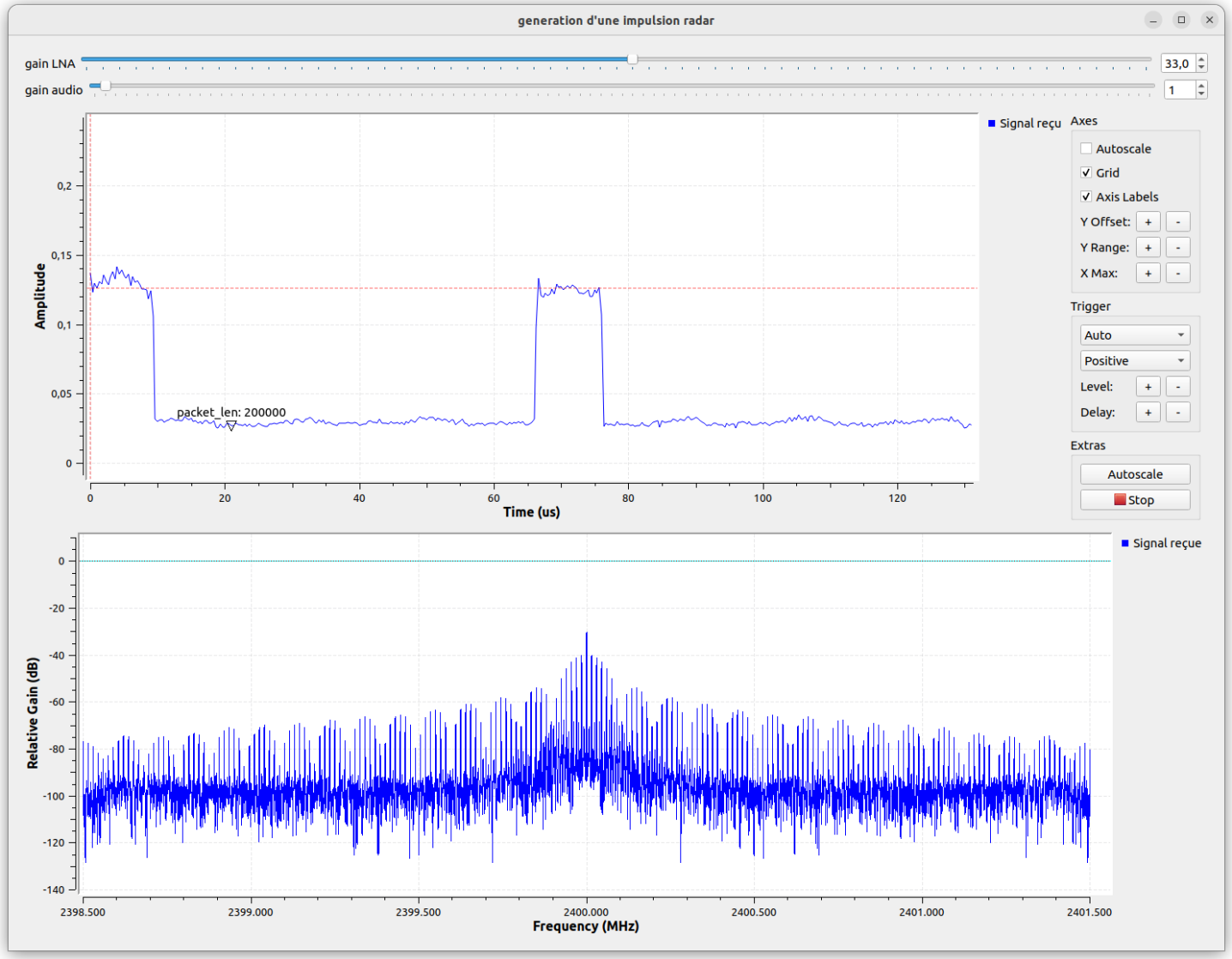
Simulateur de cibles

Cas 1 : simulateur off



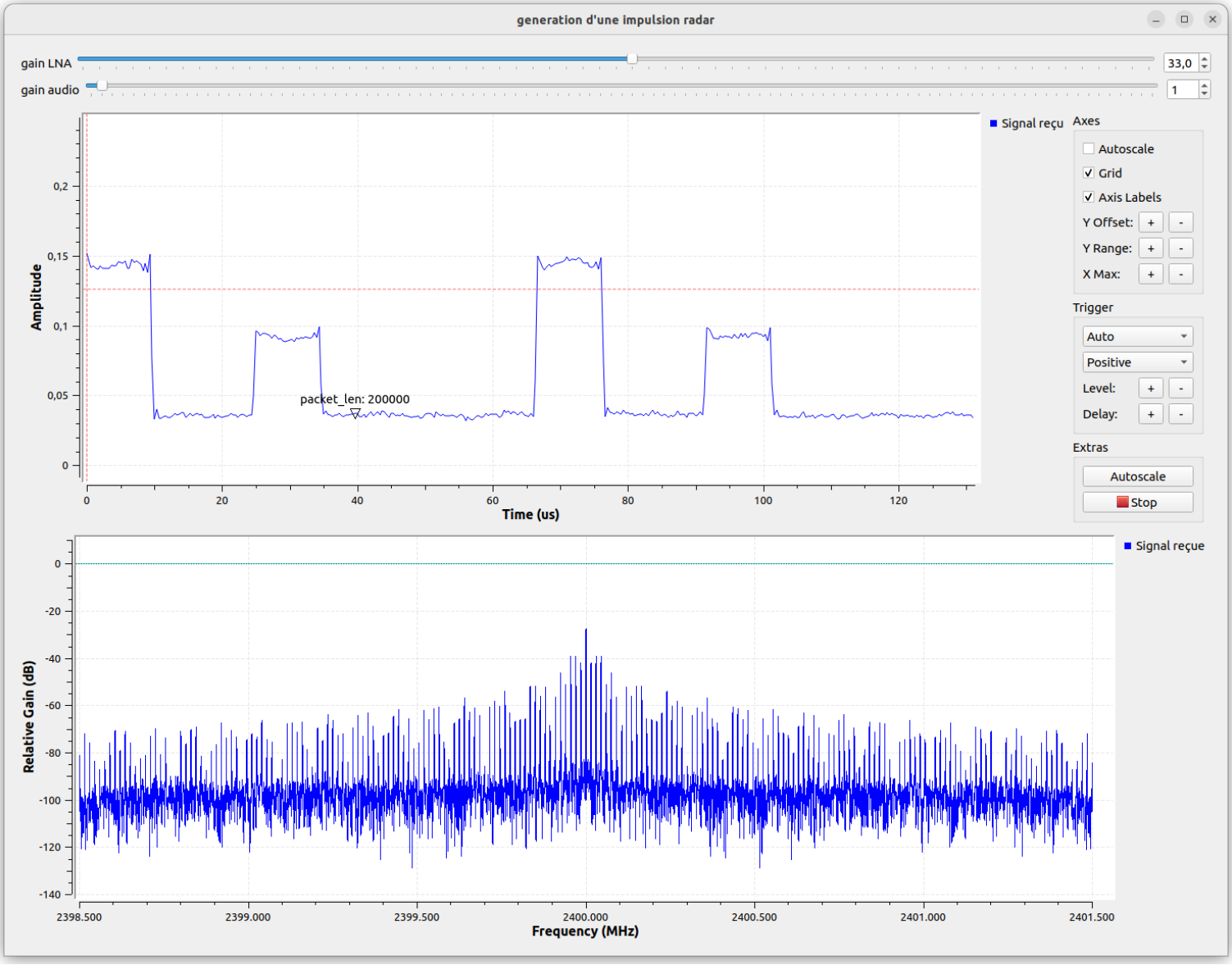
Simulateur de cibles

Cas 1 : simulateur on – pas de cible



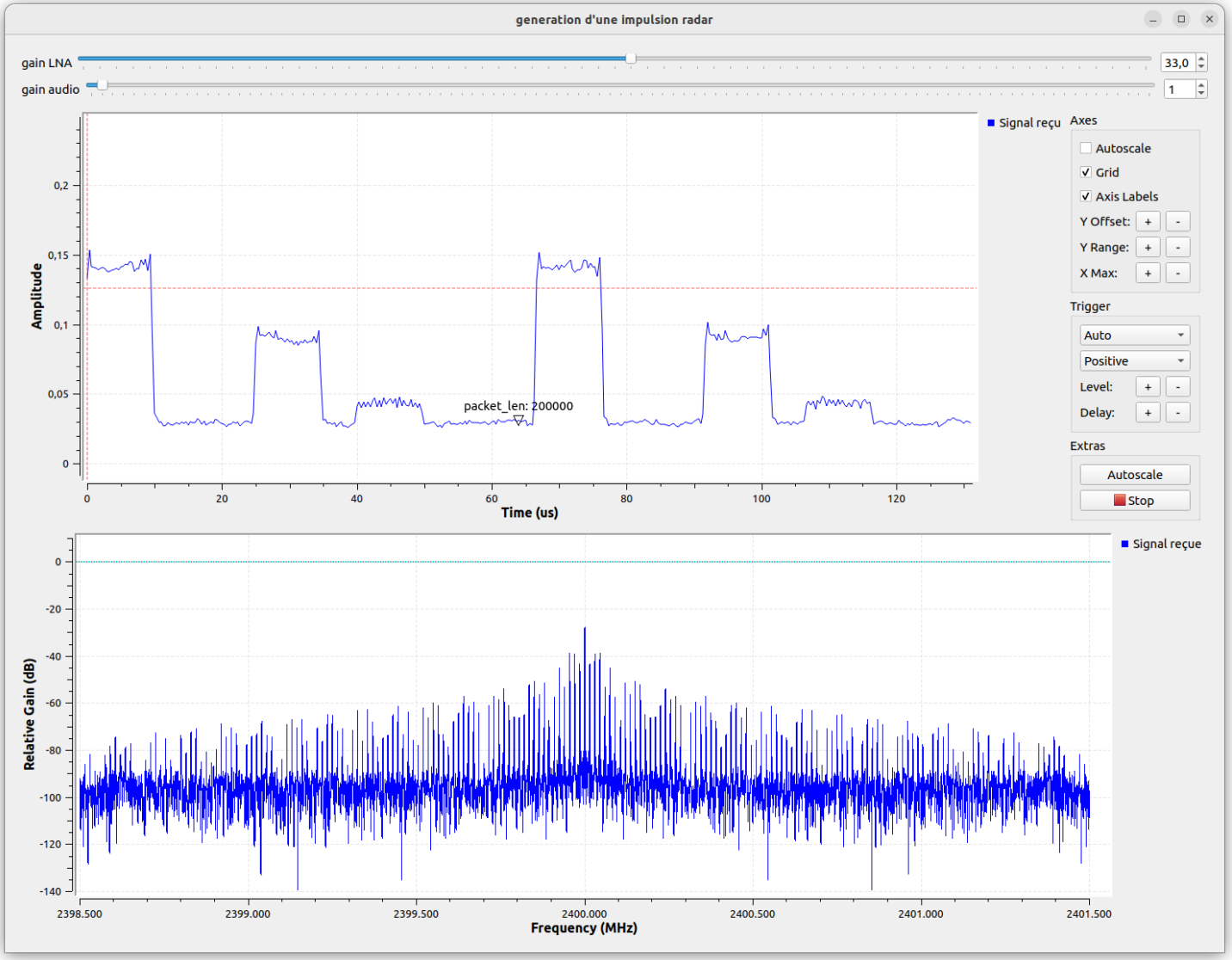
Simulateur de cibles

Cas 2 : une cible



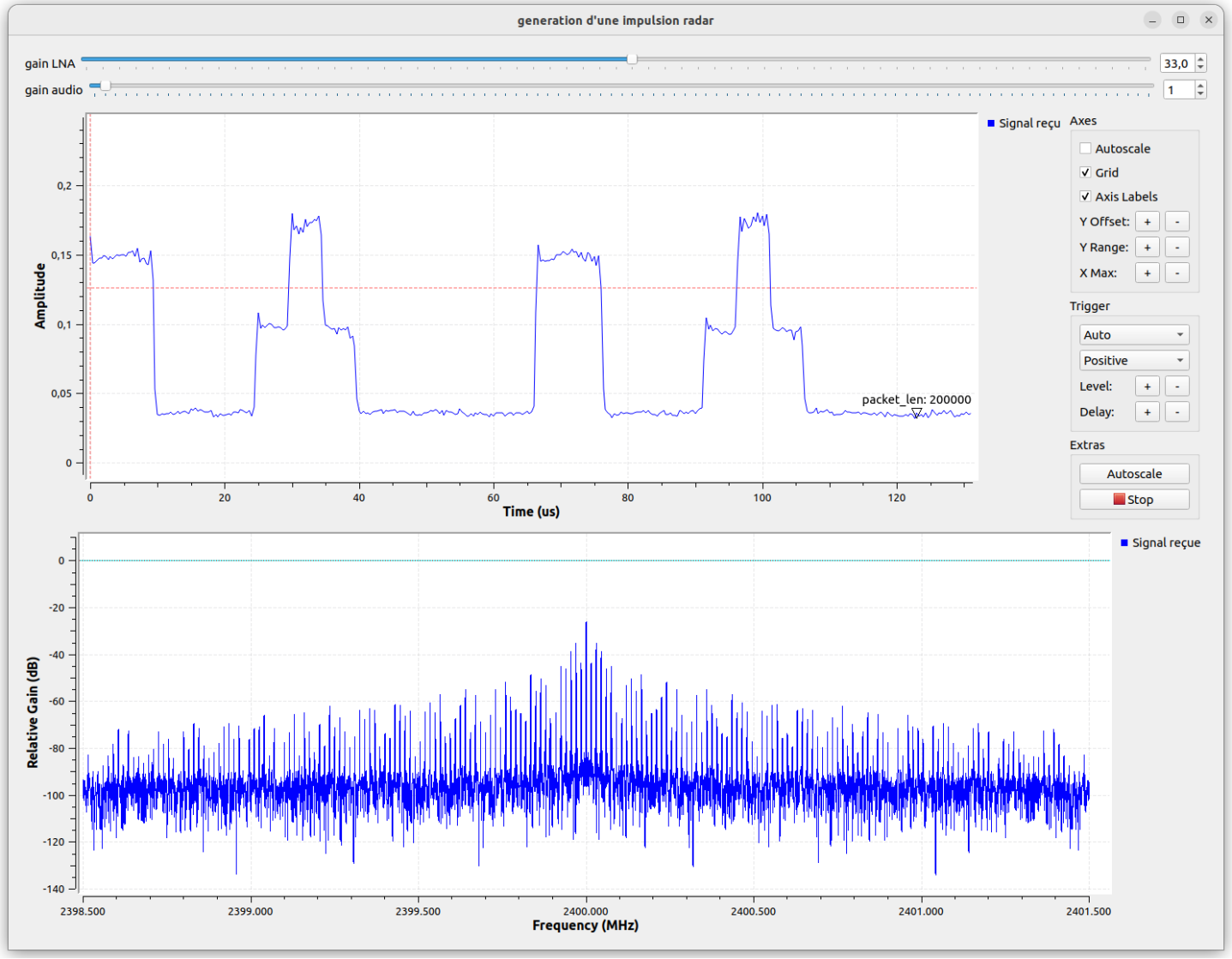
Simulateur de cibles

Cas 3 : deux cibles



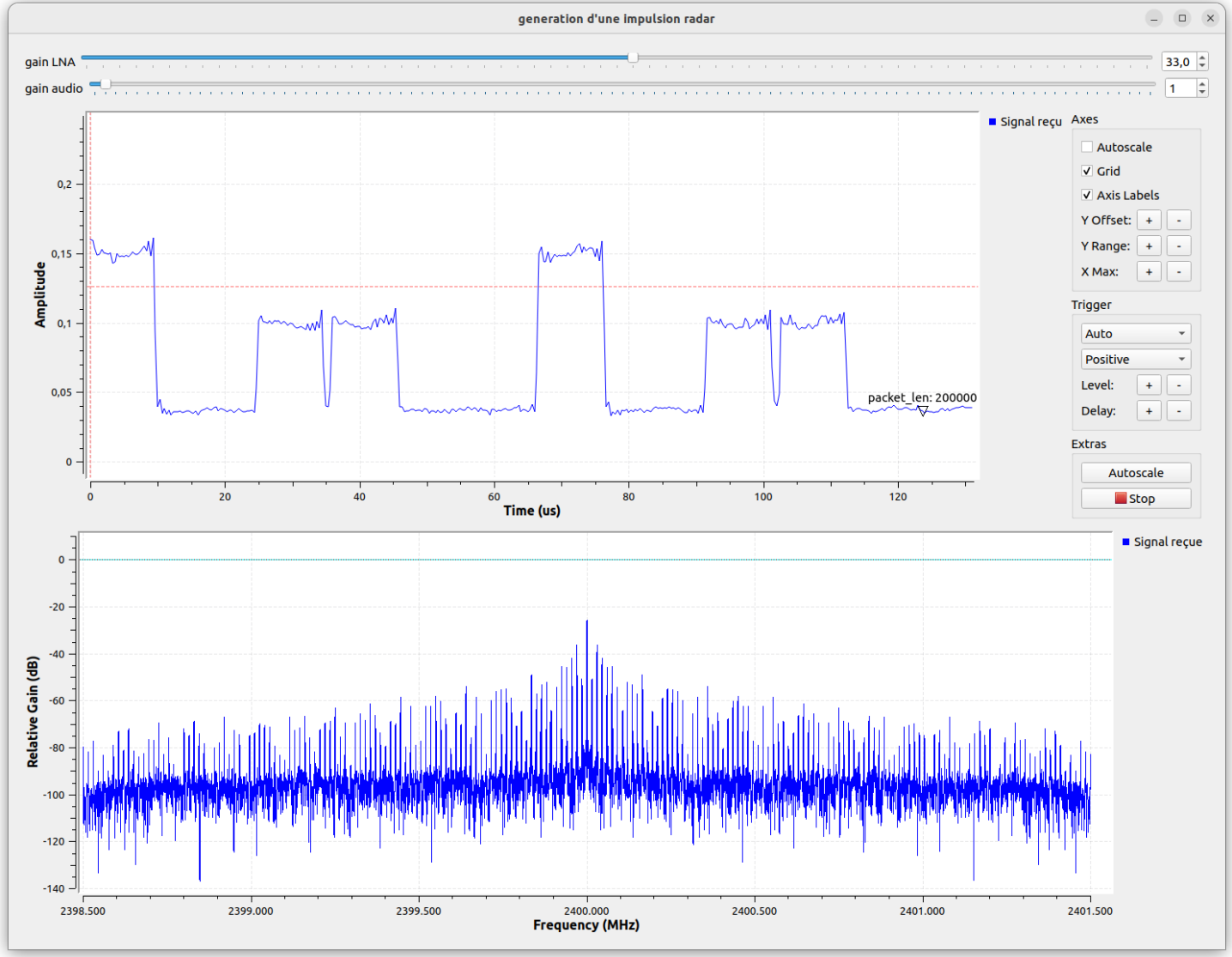
Simulateur de cibles

Cas 4 : deux cibles - résolution



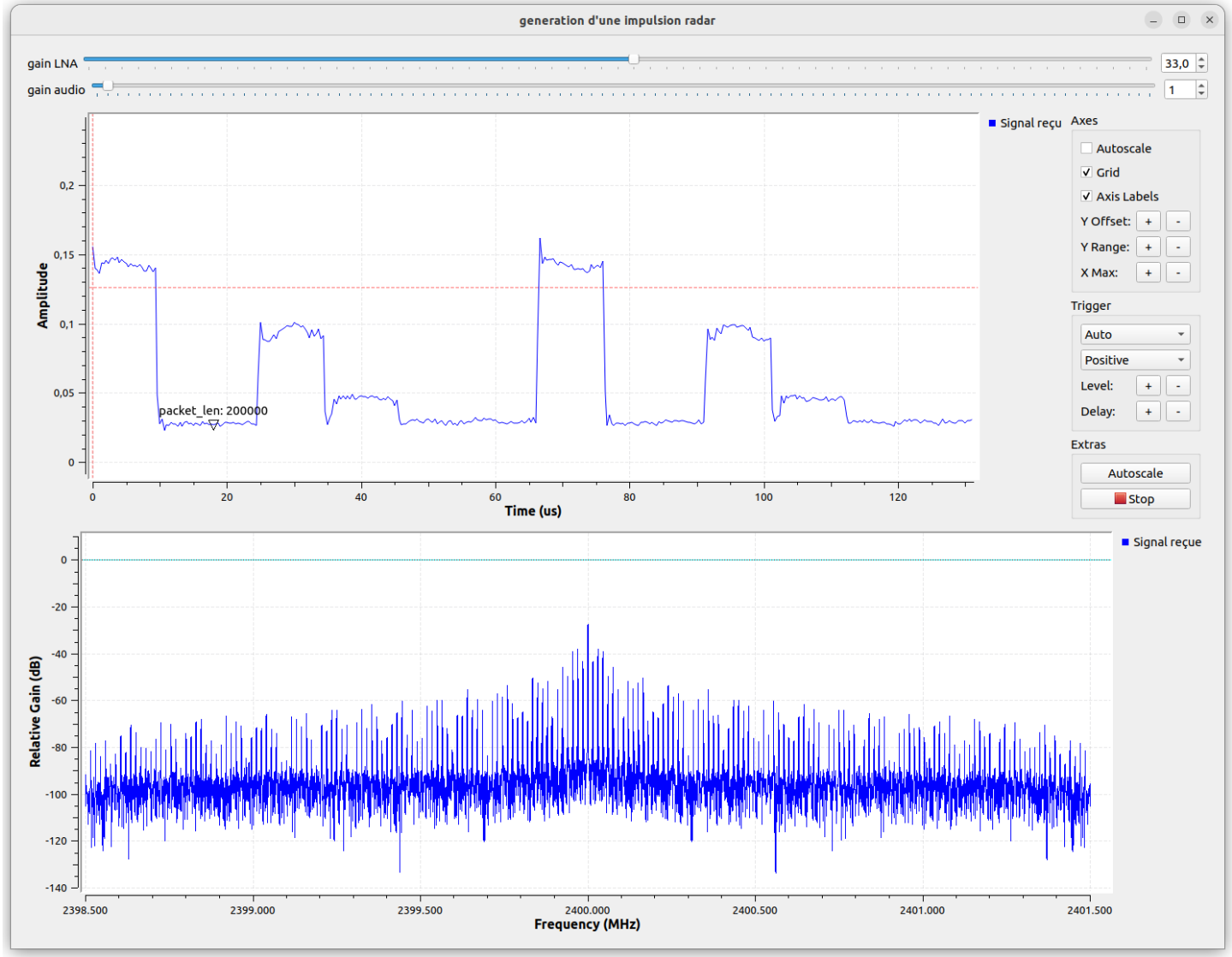
Simulateur de cibles

Cas 4 : deux cibles - résolution



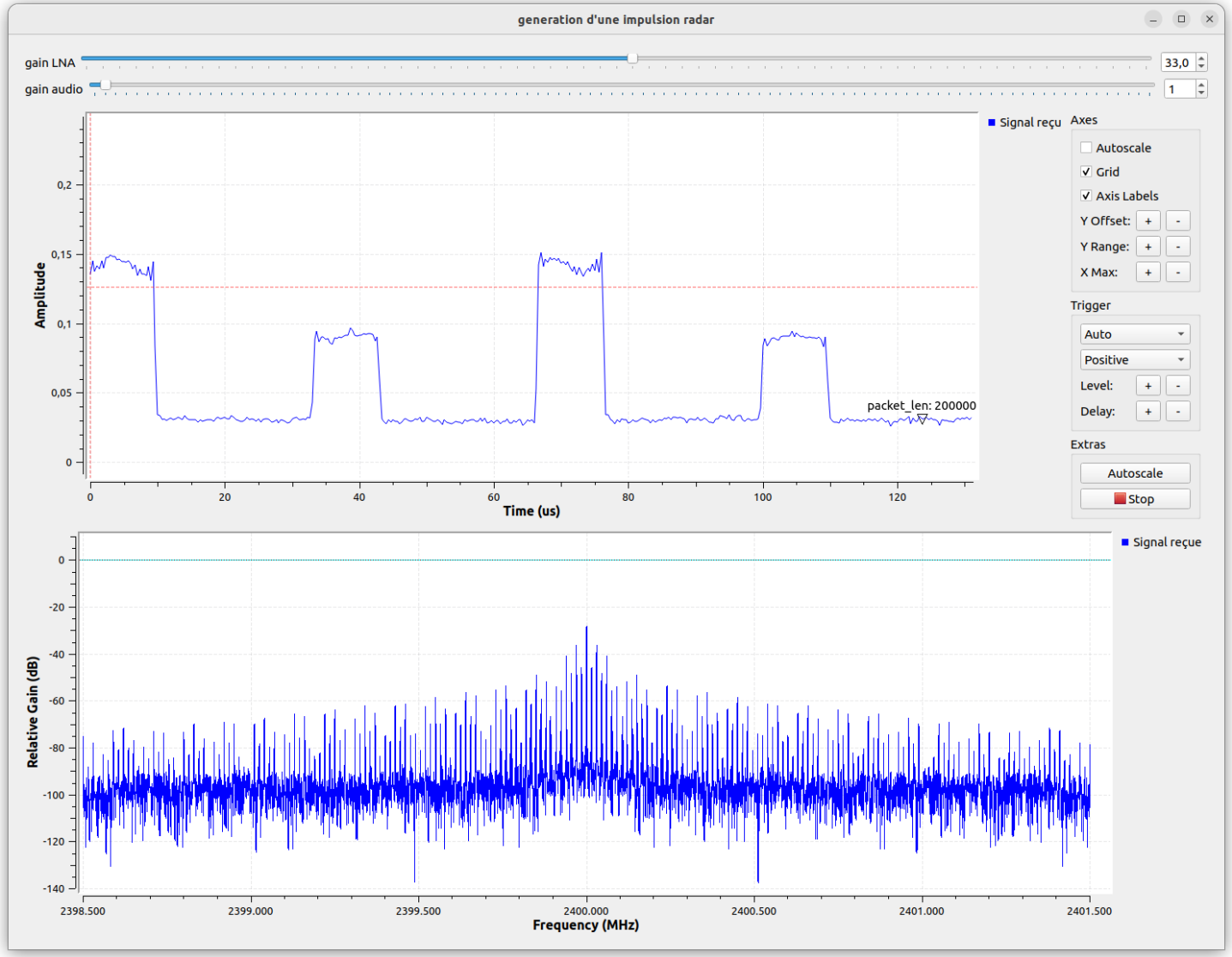
Simulateur de cibles

Cas 4 : deux cibles - résolution



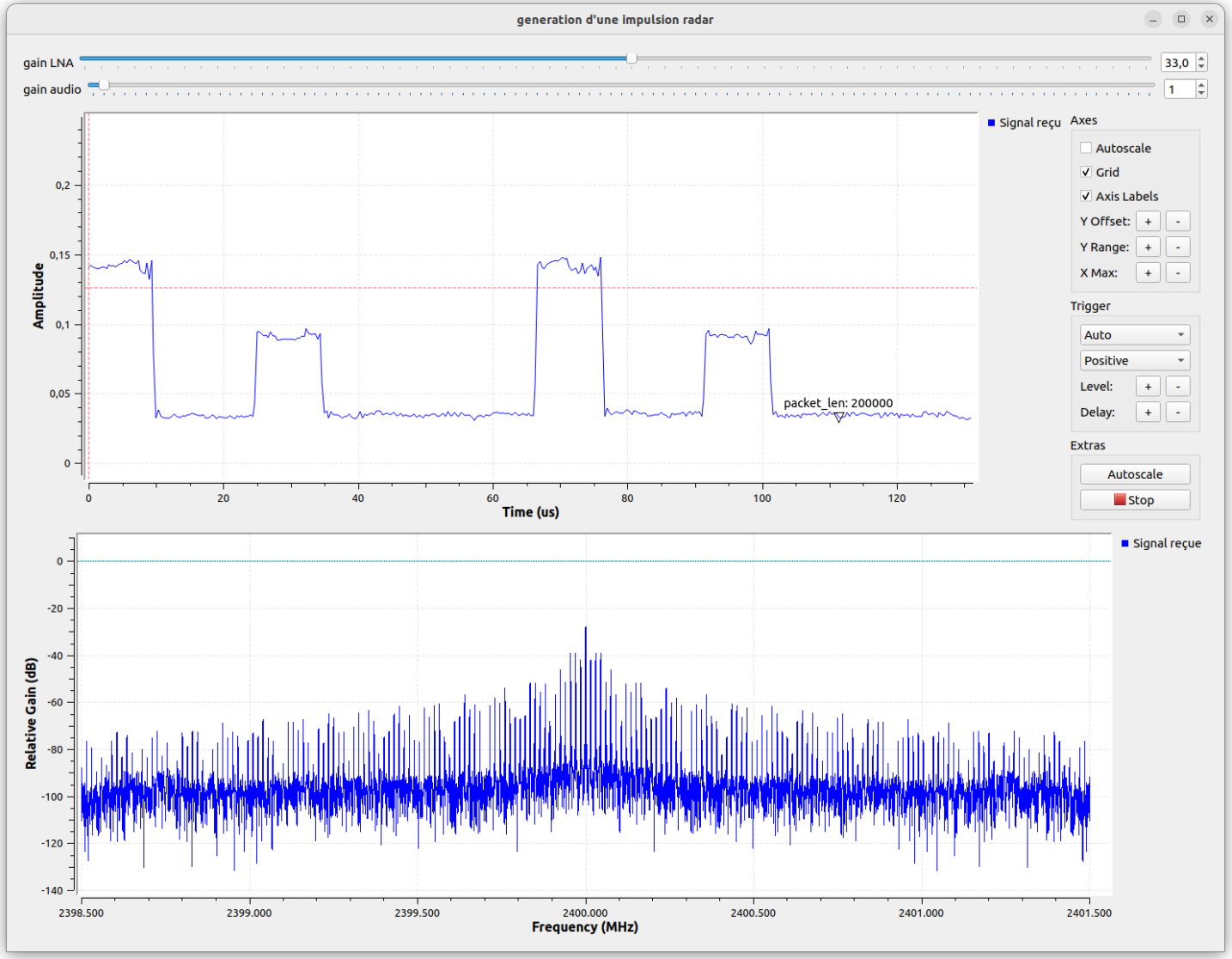
Simulateur de cibles

Cas 5 : ambiguité (tp = 100us)



Simulateur de cibles

Cas 6 : effet du doppler - 70Hz



Simulateur de cibles

Cas 6 : effet du doppler - 500Hz

